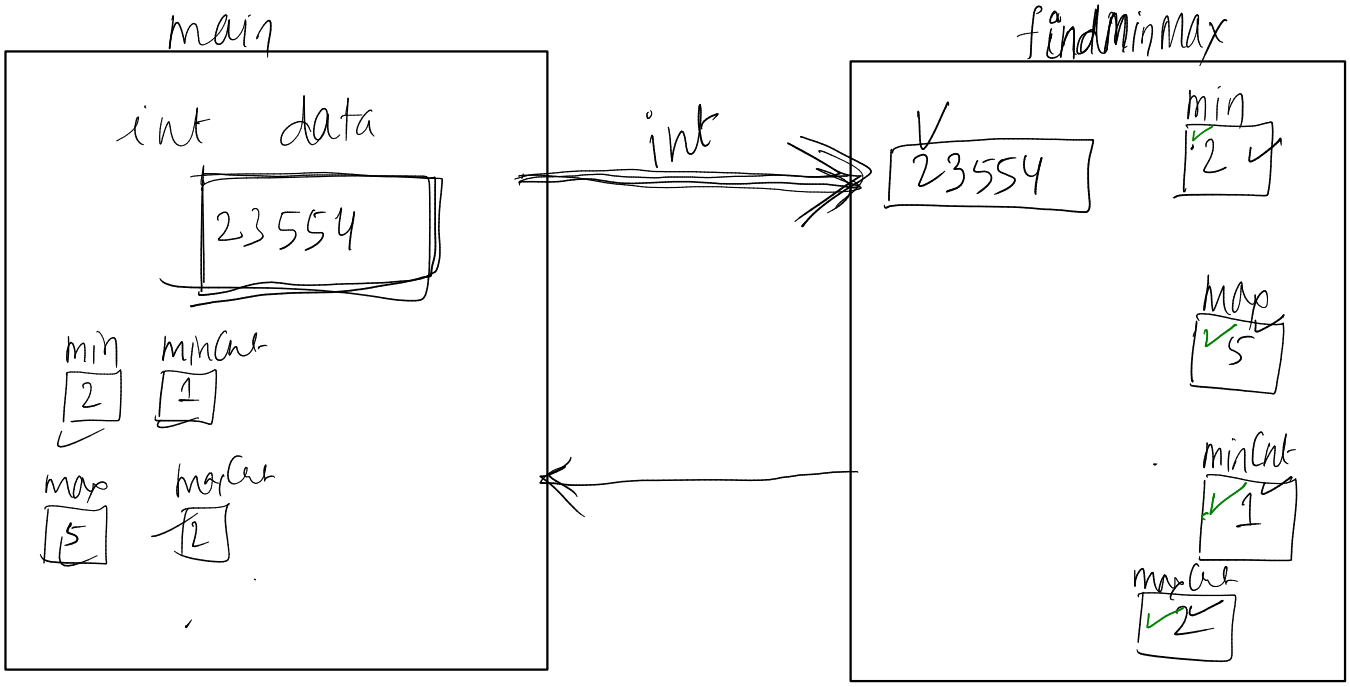
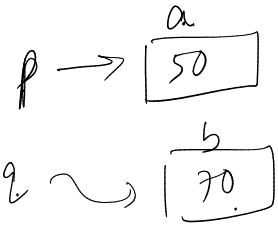


NOTE: Printing done in main only
computation is NOT in main

How to return multiple data from a function?

- structure (if data are different type)
- Array (if data are same type)

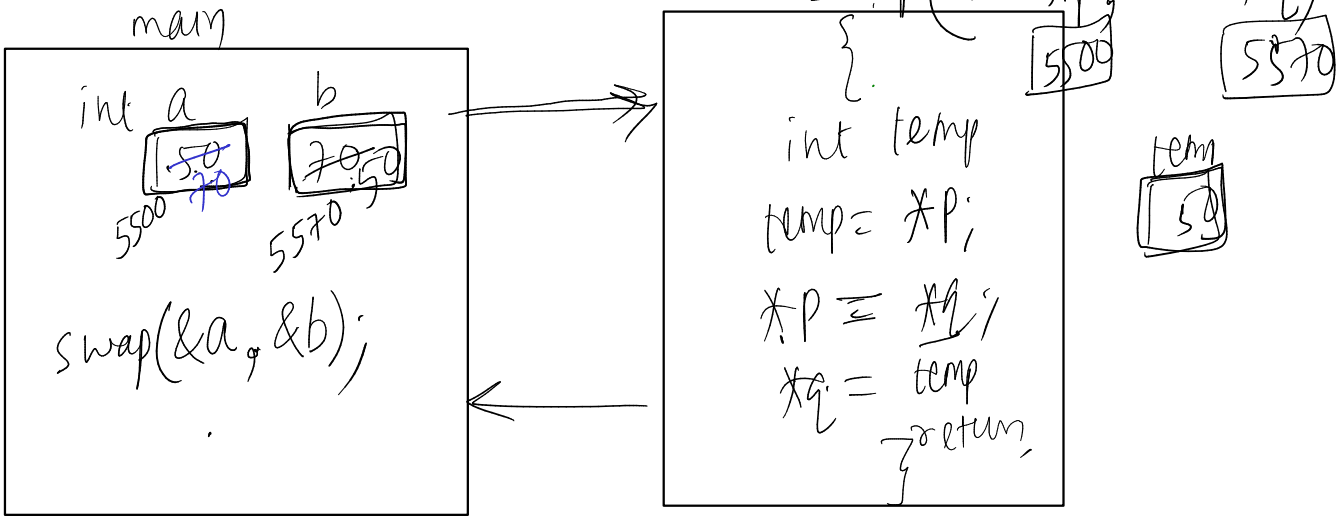
⇒ Call by reference.



Call by Reference

→ pass the address of variable

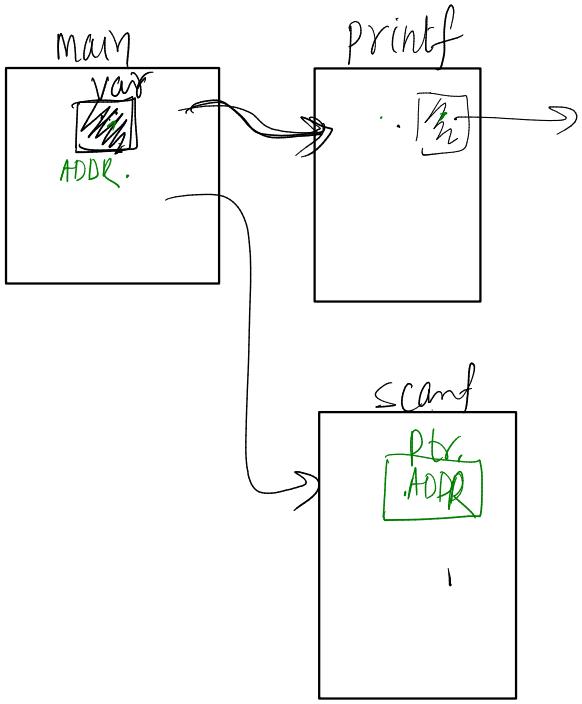
collection by Pointer



printf("%d", n);

Q. Why scanf needs &(Address) whereas printf doesn't.

sf("%d", &n);




```
main() {
    findmax(data, &min, &minCnt, &max, &maxCnt);
}
```

```
findMinMax(int var, int *pmin, int *pMinCnt, int *max, int *maxCnt)
{
int min=9, max=0, minCnt=0, maxCnt=0;
for( temp = var ; temp ; temp /=10)
  {  if (temp%10 < min)  min = temp%10;
     if (temp%10 > max)  max = temp%10; }
for(temp=var ; temp ; temp /=10) {if (temp%10 == min)
                                    minCnt++;
                                  if(temp%10 == max) maxCnt++;
                                  }
*pmin = min;   *pMinCnt = minCnt;
*pMax = max;   *pMaxCnt = maxCnt;
}
```

void *, float *vs char *, NULL Pointer, const * DECAY

ARRAY

→ User defined Datatype,

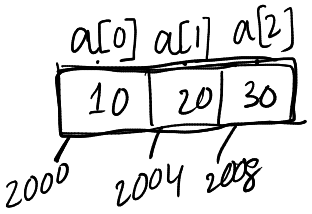
→ collection of elements of same type, in adjacent Memory.

Syntax: datatype arr_name[size];

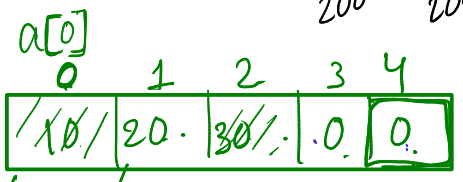
e.g. char name[20]; // name is an array of 20 characters

int arr[10]; // arr is an Array of 10 integers.

```
main(){
//int a[.]; // Error;.
int a[] = {10, 20, 30}; // valid
```



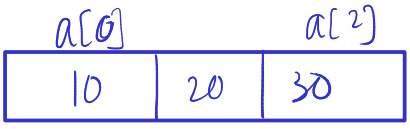
int a[5] = {10, 20, 30};

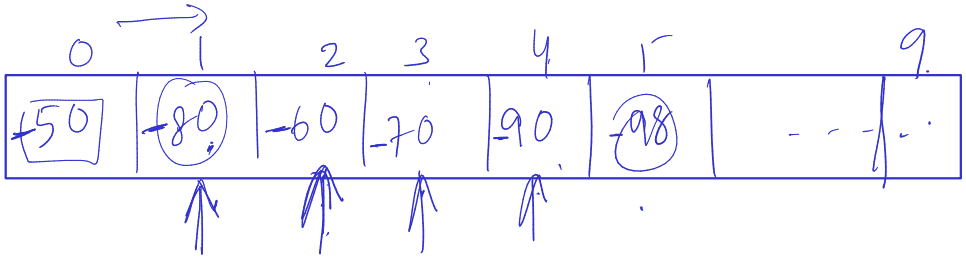


// Partially initialized Array,

the remaining elements are filled with zero

int a[3] = {10, 20, 30, 40, 50};





traverse the Array to find the Higher data

MAX = ~~50~~ = a[0];

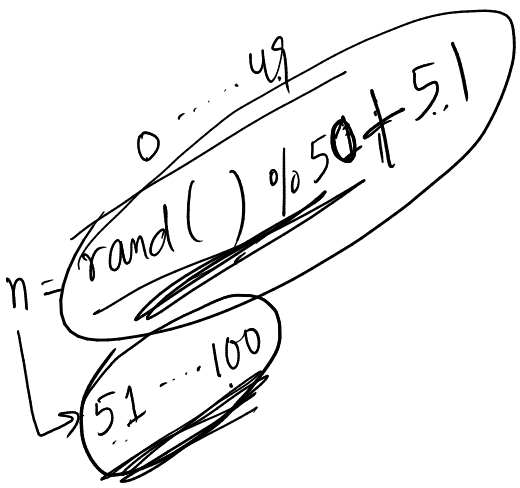
50
~~80~~
~~90~~
98

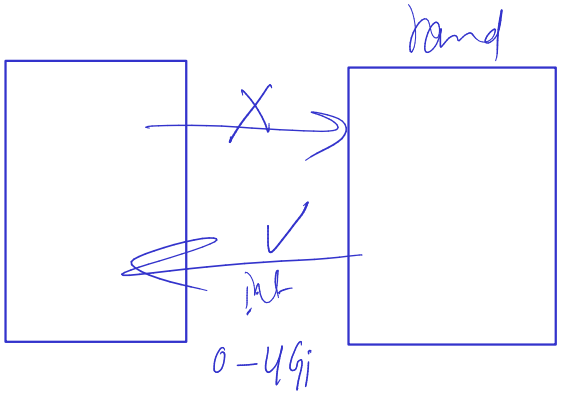
```
for(i=1; i<10; i++)
{
    if(arr[i] > max)
        max = arr[i];
}
```

→ `pf("The ter : %d", max);`

`int rand(void);`

`n = rand();`





`n = rand() % 5;`
↳ 0 4

1 6

`n = rand() % 6 + 1`
0 ..... 5 + 1
↳ 1 6

W.A.P to count the vowels in an Array of 20 alphabets.

I

→ char arr[20];

→ Use rand() to generate the UPPER case Alphabets

II In the above program make changes to

→

→ generate the alphabets (combination of upper & lower)

III In the above program, make sure that all data are Unique